Predicting Road Accident Severity Using Machine Learning: A Global Perspective

Overview:

Road traffic accidents indeed operate as meritorious public safety issues across the entire globe, costing approximately 1.3 million lives every year and resulting in heavy economic and healthcare burdens. Machine learning methods have been studied as a means of predicting road accident severity with a real-world set comprising about 132,000 entries spanning many countries. Features include type of road, weather condition, traffic volume, visibility, and response time for emergencies. Preprocessing of data comprised encoding, scaling, and enforcing class balance (using SMOTE) for maximum performance of models. Five classification models, Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM, were trained and tested in severity-prediction feasibility. Advanced modeling notwithstanding, the models scored accuracies like a hundredth, placing them around 33%, which apparently sets some severe limitations on the dataset in distinguishing levels of severity. Emphasis is laid by the study on improving future predictive accuracy through more granular features or through external data sources.

Summary:

The project is set about predicting the severity of a road accident due to a global dataset carrying about 132,000 records. After some preprocessing and class balancing using SMOTE, various classification algorithms — LightGBM, Random Forest, and XGBoost — were then trained and tested. Even with advanced models and fine-tuning, all methods provided comparable results (~33% accuracy), signifying that the given features might not be strongly predictive by themselves. These results indicate that the richer data or domain-specific features are vital in the actual severity forecast.

Project Description:

Road traffic accidents are a leading cause of injury and death worldwide, with more than 1.3 million deaths and tens of millions of injuries reported each year according to the World Health Organization. Such incidents not only cause significant human suffering but also impose enormous financial costs on affected individuals, their families, governments, and healthcare systems. Prediction of road accident severity could bring forth a great change in public safety issues.

If severity can be predicted accurately, emergency services can prioritize dispatch efforts, policymakers can build safer infrastructure, and insurance companies can better model risk. Machine learning becomes a powerful tool to garner actionable insights and develop predictive

models with large-scale road accident datasets being available and data having been collected from different sources.

The project is launched because of the opportunity to utilize the strengths of machine learning techniques to investigate whether information currently existing in road accident datasets can be meaningfully exploited to forecast accident severity, whereby features like road, weather conditions, traffic volume, visibility, and emergency response time are all considered. It is an attempt of the present study to contribute to the expanding realm of data-based planning for public safety and road advancement.

Project Aims:

The primary goals of this project were:

- To analyze a comprehensive real-world dataset on global road traffic accidents.
- To arrive at interesting effects and understand important features influencing the severity of accidents on the road.
- To implement different supervised machine learning models to predict the severity levels of accidents.
- To address data imbalance issues through the Synthetic Minority Oversampling Technique (SMOTE).
- To evaluate models and compare across accuracy matrix, confusion matrix, and ROC.
- To analyze results to find significant predictive features.
- To take note of the weaknesses of the existing dataset and suggest how to further increase it.
- To provide insights that could assist in transportation planning, emergency response, and insurance risk assessment.

Dataset Description:

The dataset used for this project had its collection of approximately 132,000 anonymized road traffic accident records from Kaggle. Some features considered in the dataset include the road type, weather conditions, traffic volume, visibility, emergency response time, and casualty count. In addition, the data contains categorical and numerical attributes that may have been preprocessed using one-hot encoding and min-max scaling. To overcome this imbalance of class in road accident severity, the SMOTE technique of data sampling was applied on the dataset. The resultant cleansed data were then used for training and evaluation of numerous classification models.

Methods:

1.Exploratory Data Analysis (EDA):

The first step was Exploratory Data Analysis to understand the distribution and relationship among variables relating to accident severity. Numerical features such as Visibility Level, Traffic Volume, Speed Limit, and Emergency Response Time were visualized with histograms and boxplots to identify outliers, skewness, or variance. Categorical features like Weather Conditions, Road Type, and Day of-the-Week were assessed through count plots to check if different categories suggest variations in accident severity. This allowed deciding upon useful features and new variable engineering. Also, a significant class imbalance was observed for the target variable (severity), which influenced further modeling steps.

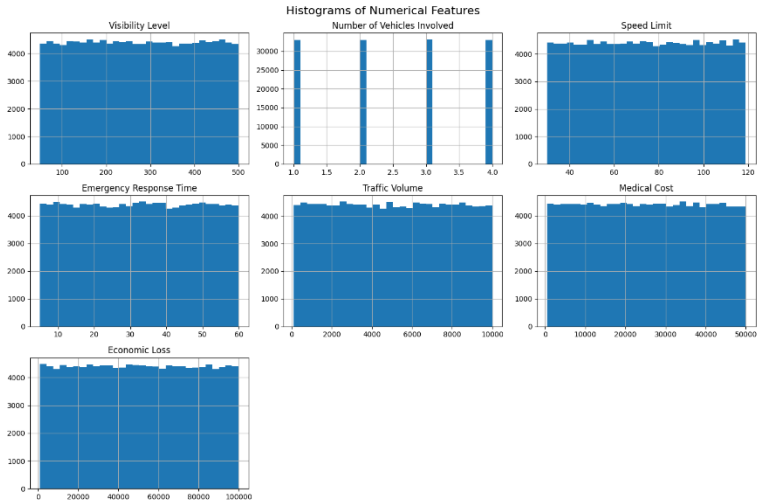

Fig 1: Histograms of Numerical Features

This plot shows the distribution of key numerical variables in the dataset. Most features like Visibility Level, Speed Limit, and Emergency Response Time appear uniformly distributed. The variable Number of Vehicles Involved shows clear spikes at specific counts, indicating categorical-like behavior despite being numeric.

2.Data Preprocessing:

Prior to model training, the preprocessing of the dataset was carried out to maintain consistency and make it compatible with machine-learning algorithms. Categorical variables like Weather Condition, Road Type, and Day of the Week were one-hot encoded. Continuous variables such as Emergency Response Time, Traffic Volume, and Medical Cost were scaled using Min-Max normalization, thus confining their values within the 0 and 1 range. Missing values were imputed

for non-critical fields where feasible, and cases with missing target labels or critical data were discarded.

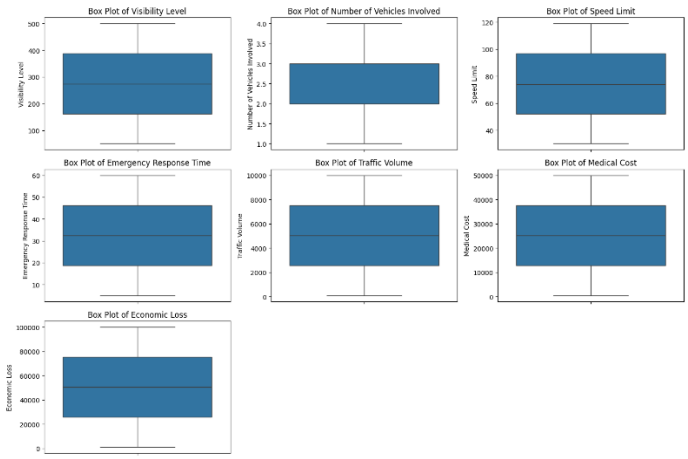

Fig 2: Boxplots of Numerical Features

This visual display displays boxplots for key numerical features in the dataset. Variables such as Speed Limit, Medical Cost, and Economic Loss exhibit noticeable variability and outliers.

3. Class Imbalance Handling:

One major challenge in this dataset was the imbalance of the target variable, Accident Severity. Most of the records belong to the "Minor" label, while the representation of the "Severe" class was very low. Such an imbalance may bias the model during training and cause it to predict the minority classes with poor performance. To overcome this imbalance, SMOTE was employed on the training set. SMOTE works by creating synthetic examples from minority classes by interpolating between existing minority class samples, thereby introducing balance into the data and giving the model a fair chance during learning.

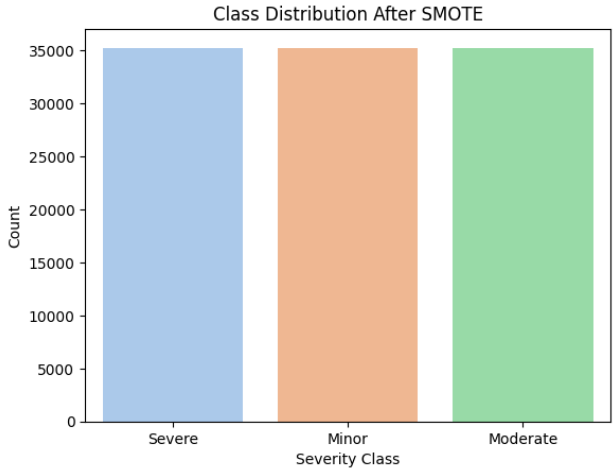

Fig 3: Class Distribution after SMOTE

4.Model Training:

After some preprocessing, which included class balancing, five supervised machine learning classification algorithms were used to estimate accident severity:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier
- LightGBM Classifier

The dataset underwent an 80/20 ratio split for training and testing. SMOTE was applied on the training set only to avoid data leakage occurrence. In the first trial, the models were trained using default hyperparameters, following which key parameters (such as tree depth and number of estimators) of Random Forest and LightGBM were tuned manually to attain higher performance.

Models were trained through scikit-learn and XGBoost/LightGBM libraries. Hyperparameters for Random Forest and LightGBM have been tuned manually:

- Random Forest: n_estimators, max_depth, and min_samples_split
- LightGBM: learning_rate, num_leaves, and n_estimators

The target classification was multiclass (Minor, Moderate, Severe), which involved an added layer of complexity for evaluation. Therefore, the models were evaluated based on:

- Accuracy
- Precision, Recall, and F1-score (macro-average)
- Confusion Matrix
- Multiclass ROC-AUC Score

After all the tweaking and balancing, all the models came out with a comparable overall accuracy of around 33%, indicating there is very little of-able separation between the severity classes with the features used presently. However, the evaluation of models shed light on what the most significant features are and heavily where the models cannot distinguish well between “Moderate” and “Severe” cases.

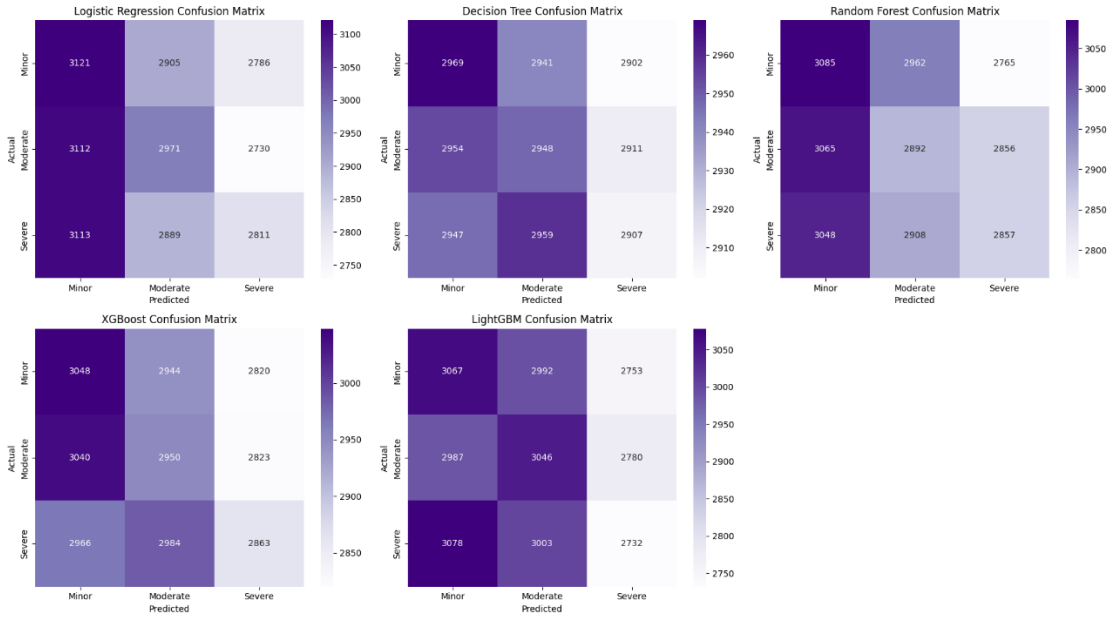

Fig 4: Confusion Matrix of All Models

5.Model Evaluation:

Model performance was gauged by evaluating a comprehensive set of measures well-adapted for multiclass classification tasks. Models were trained on the SMOTE-balanced training set, and performance was evaluated on the original (imbalanced) test set so as to reflect real-life performance. The metrics observed were:

- Accuracy-Overall correctness of predictions.
- Precision, Recall, F1 Score (macro-averaged)-These characterize per severity class performance of the model, with special focus on imbalanced datasets.
- Confusion Matrix-A means of establishing where misclassification occurred, by juxtaposing the actual classes against predicted classes.
- ROC-AUC Score (macro)-Appraises the ability of the model to separate classes; it is of special utility in imbalance and multiclass settings.

The five models all exhibited similar performance and with accuracy values circa 33, which means, almost random within the three levels of severity. This reflects the deficiency of the current data set in being able to effectively discriminate against the target classes with respect to features internally present.

Among the models:

- Random Forest and LightGBM had slightly better F1-scores and macro-averaged ROC-AUC.

- The confusion matrices revealed that the "Moderate" and "Severe" classes were mostly misclassified as "Minor" in all the models.

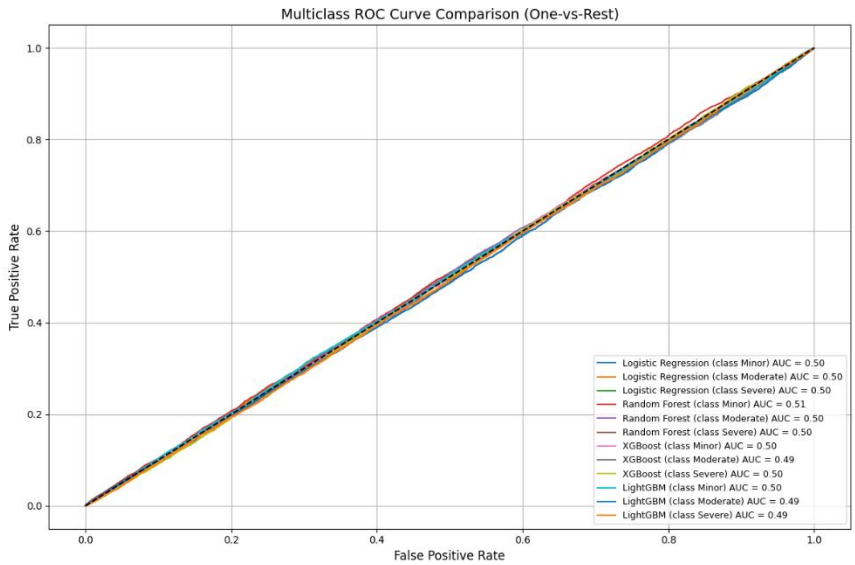

Fig 5: Multiclass ROC Curve Comparison

6.Feature Importance & Interpretation

Feature importance analysis was performed using the LightGBM classifier to interpret the models and identify what variables influenced predictions most. LightGBM is the best choice as it assigns merit to each feature based on how many times and how effectively that feature is used to reduce loss during training. The features ranked the highest by importance included **Emergency Response Time, Speed Limit, Visibility Level, Traffic Volume,** and **Medical Cost**. The presence of these variables on splits across similar decision trees hinted that these features might be at the center of the model's classification process.

While statistically important, these features did not show strong predictability on their own. For instance, Emergency Response Time and Speed Limit might contribute to model performance, but other than these two features, most features showed a significant overlap in their distributions across the different accident severity levels. This overlap created difficulties in having a model confidently distinguish between the classification of "**Moderate**" and "**Severe**" accidents. Confusion matrices also showed this ambiguity, where there were high misclassifications between the two classes. Hence, these observations suggest that the feature set may lack the complexity or depth required for multiclass prediction to be successful.

The findings bring forth a possible requirement for having more granular, context-related data. This, in turn, would help with accurate model predictions. These can be added in the form of other severity-distinguishing features such as real-time weather severity, driver behavior history, road infrastructure.

Conclusion:

To predict road traffic accidents, machine learning models were attempted to be applied using a real-world dataset comprising over 132,000 records collected from different countries. A wide assortment of features such as road type, weather condition, traffic volume, visibility level, and emergency response time was provided in the data. Everything followed a structured pipeline: exploratory data analysis and feature engineering came first, followed by preprocessing, normalization, and resolving class imbalance using SMOTE. Five of the most popular classification algorithms, namely Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM, were trained and tested to see which was best at predicting the three levels of accident severity: Minor, Moderate, and Severe.

Having applied fierce modeling methods and balancing techniques, all the models put on similar performance, hitting accuracy clusters around 33%. This signified the insufficient strength of available features to serve as obvious signals for multiclass classification. Evaluations in terms of macro-averaged precision, recall, F1-scores, and confusion matrices showed a sorely adverse impact on the models' ability to differentiate between “Moderate” and “Severe”. Advanced approaches like LightGBM and XGBoost should have been able to capture more of the subtle relationships between features and accident outcomes- but they didn't! Further analysis of feature importance revealed that, although emergency response time and speed limit were the most influential factors, by themselves, they could not adequately differentiate severity levels.

From these findings, several avenues for future work emerge. Great improvements can be made to the dataset, in providing finer-grained and more contextual data. This may include geographic information (accident location, population density, etc.); environmental variables (weather data in real-time, road surface condition, etc.); vehicle attributes (vehicle type, speed at impact); human behavior data (driver's condition, distraction indicators). Future work may be extended to examples of ensemble stacking, deep learning models, and feature interaction modeling to capture complicated interaction patterns in the data. In the end, providing higher accuracies for this modelling solution would therefore direct emergency response systems, urban planning, and risk assessment tools both governmental and private to be smarter.